Graph-Enriched Lakehouse

Combining Databricks Genie with Neo4j Graph Data Science

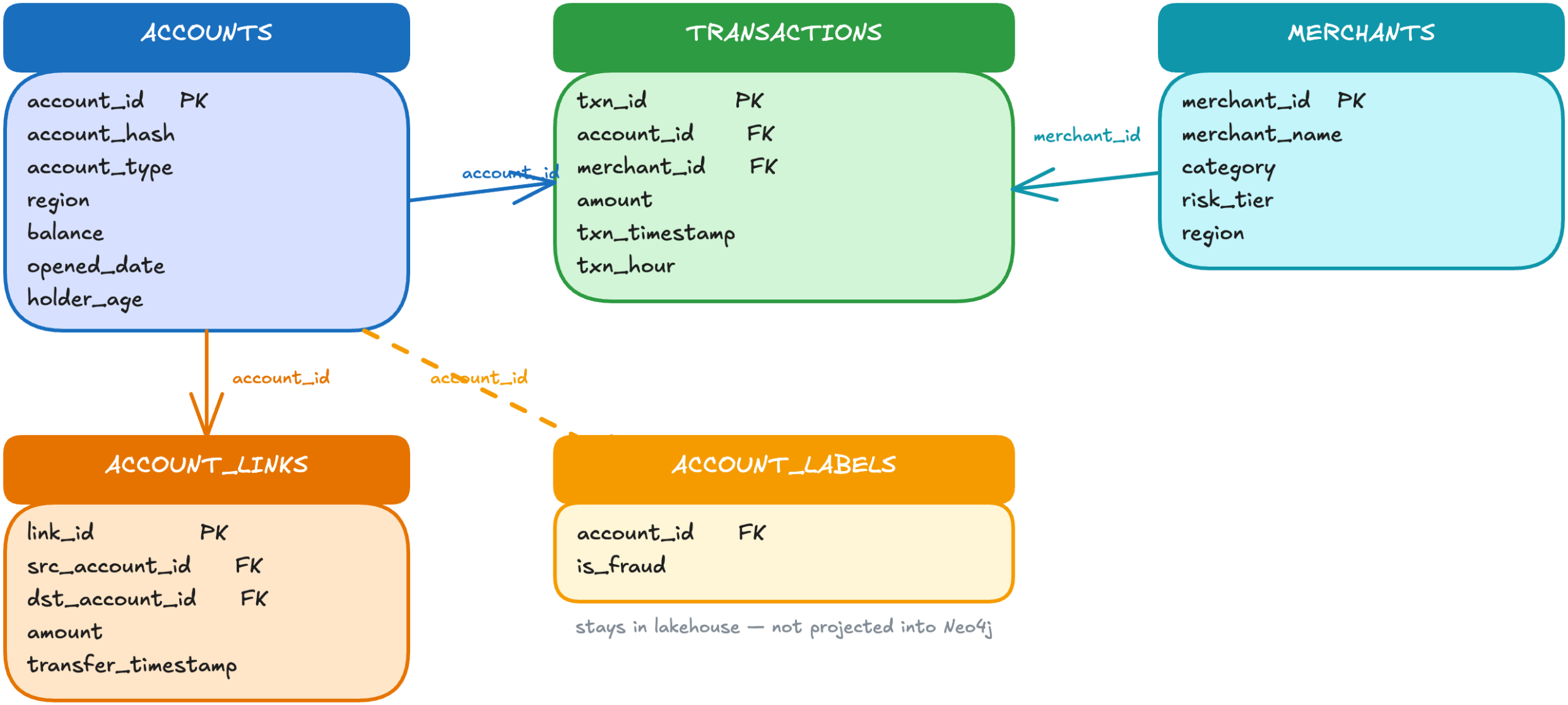
Financial Crime Hides Between the Rows

- **Coordinated fraud** spreads activity across dozens of accounts on purpose
- **Each transaction looks clean in isolation**
- **The fraud is hidden in the patterns across the accounts**
- **Row-level aggregation cannot produce a property that lives in the connections**

Demo Data Set: Synthetic Banking Network

- **Two financial networks:** a spending graph and a peer-to-peer transfer graph
- **Spending graph:** accounts transact with merchants
- **Transfer graph:** accounts send money directly to other accounts
- **Fraud rings** leave structural footprints in both
- The goal: surface those patterns, not score individual transactions

Data Model — Silver Layer



25,000 accounts · 7,500 merchants · 250,000 transactions · 300,000 account_links

Genie: Merchant Favorites (Before Graph Enrichment)

"Which merchants are most commonly transacted with by the top 10% of accounts by total dollar amount spent across merchants?"

Merchant	Transactions
MarketPlus	30
BeanStreet	28
GroceryHub	27
QuickCash	26
FuelZone	26

Top five tied at 26-30. No merchant stands out for investigation.

Genie: Merchant Favorites (After Graph Enrichment)

"Which merchants are most commonly transacted with by accounts in ring-candidate communities?"

Merchant	Top-decile (before)	Ring members (after)
MarketPlus	30	**111**
QuickCash	26	**104**
CoinVault	not ranked	**101**
LoanEdge	not ranked	**99**

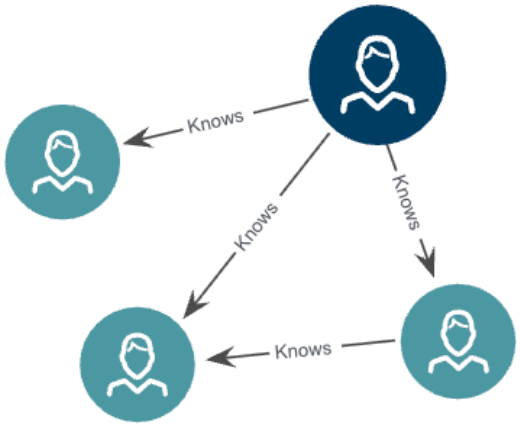
5% of accounts. 4× the transactions. Per account: 7×.

Volume proxy: where everyone goes. Ring filter: where ring candidates concentrate.

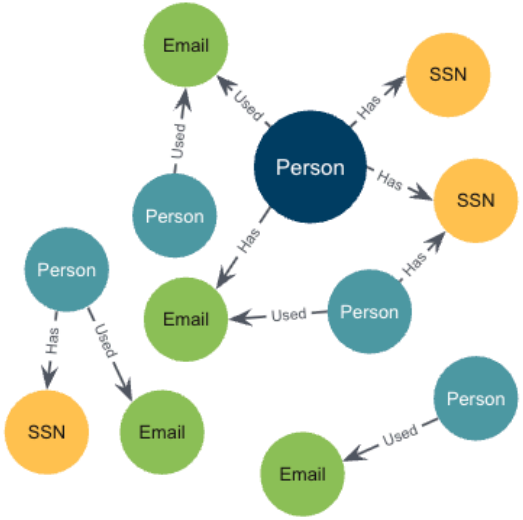
Better Data for Better Genie Answers

Graph makes it easy to explore hidden patterns

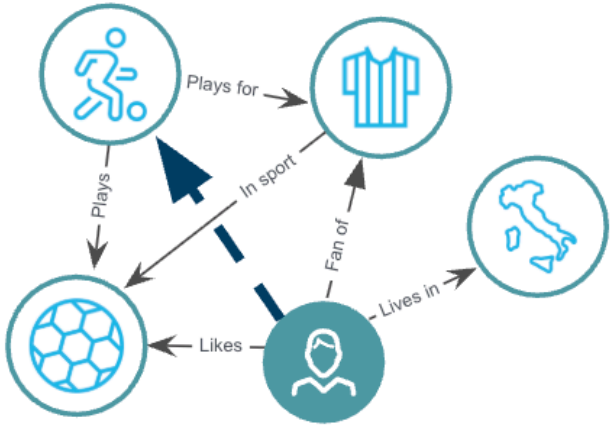
What's important?



What's unusual?



What's next?



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Graph Databases Find Every Instance of a Pattern

- **Describe a pattern:** cluster of accounts, shared counterparties, any network shape
- **Get every instance back:** no starting account, no ID to look up
- **Pattern-matching is the graph's native operation**

Better Data for Better Genie Answers

- **Graph results enrich the Gold tables:** `risk_score`, `community_id`, `similarity_score`, `fraud_risk_tier`
- **Genie treats them like any other dimension:** `GROUP BY fraud_risk_tier`, `WHERE is_ring_candidate = true`
- **New questions unlocked:** candidate-population sizing, regional review workload, merchant concentration by community
- **Change the columns. Change what Genie finds.**

What Graph Analysis Adds to Genie

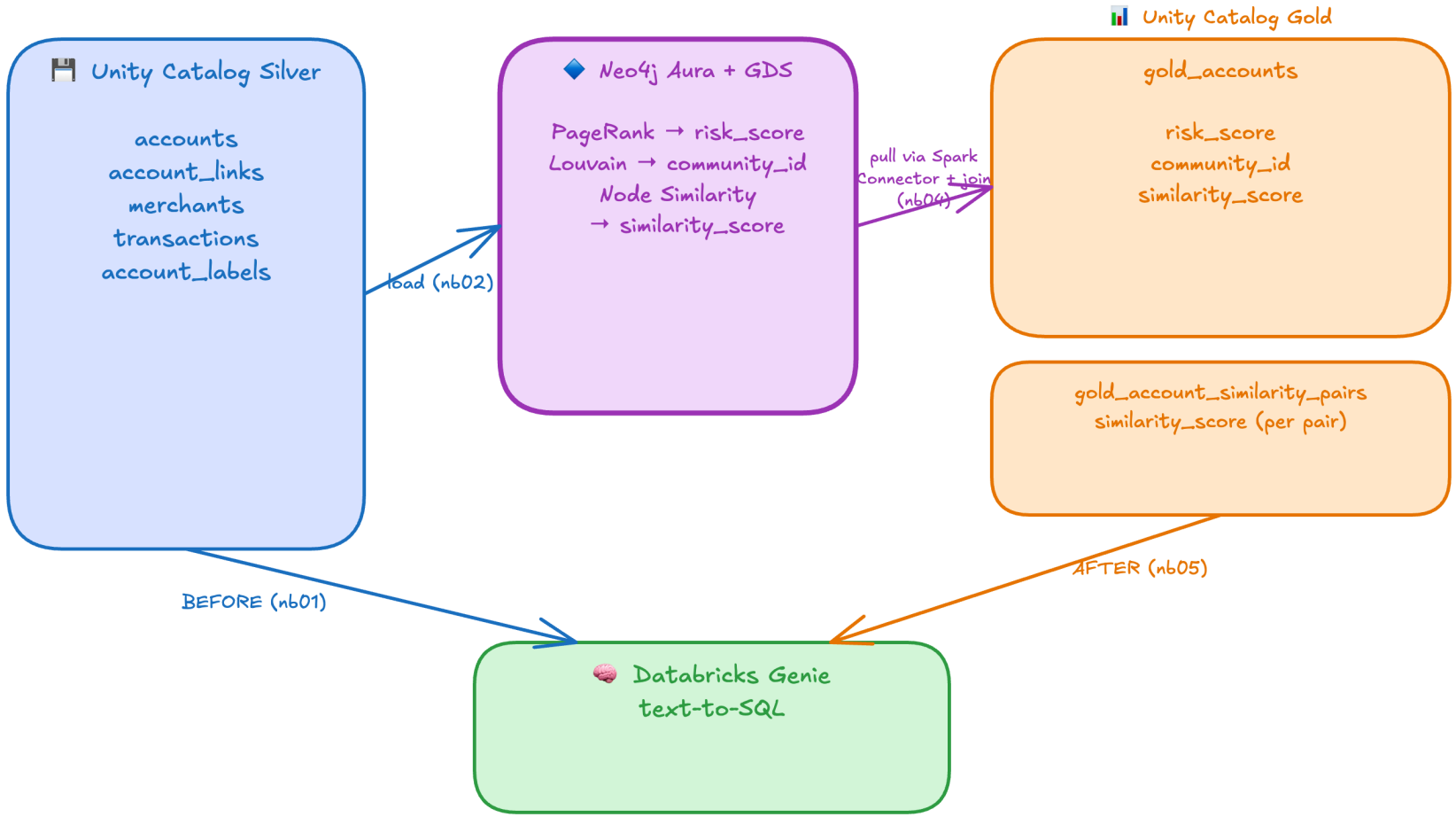
Examples of enrichment:

- **Centrality:** how central an account is in the flow of money. A network position.
- **Community:** which accounts cluster tightly together. A density across many edges.
- **Similarity:** which accounts route through the same counterparties. A neighborhood property.

Each becomes a column Genie can group by, filter on, and compare across.

The Enrichment Pipeline

Pipeline Architecture — Graph-Enriched Lakehouse



The Enrichment Pipeline

Four steps convert a network of account relationships into plain columns that Genie queries like any other dimension.

- **Load:** Silver tables into Neo4j Aura as a property graph
- **Run GDS:** PageRank, Louvain, Node Similarity against the graph
- **Enrich:** pull scores via Neo4j Spark Connector, join with Silver, write to Gold
- **Query:** enriched columns sit alongside all base data in the Gold table

The Graph Finds Candidates. The Analyst Finds Fraud.

- **GDS produces structural signals that indicate ring-like behavior:** community membership, centrality, similarity
- **A high-risk community is a candidate, not a verdict**
- **The analyst runs ordinary Genie queries against the enriched Gold tables** to decide which accounts and merchants warrant investigator time

The Analyst's Toolkit, Expanded

- `community_id` and `fraud_risk_tier` sit alongside region, product, and balance as ordinary dimensions
- **New questions unlocked:** candidate-population sizing, regional review workload, merchant concentration by community
- `GROUP BY fraud_risk_tier`, not "find the ring"

Same Genie. More answers.

Where This Pattern Applies

- **Fraud-ring surfacing:** tight-community trading, shared merchant preferences that do not fit the background distribution
- **Entity resolution:** collapsing customer, device, and household records that refer to the same real-world entity based on shared attributes and topology
- **Supplier-network risk:** tiers of supplier exposure, single points of failure, concentrations of risk in multi-tier supply graphs
- **Recommendation structure:** communities of users, products, or content with shared consumption patterns as features for downstream recommenders
- **Compliance network review:** counterparty clusters and beneficial-ownership paths that require human review under regulatory frameworks

Key Takeaways

- **One question, two answers:** the gap is the whole argument
- **Graph for connection questions, lakehouse for everything Genie already does well**
- **GDS columns land in Gold as ordinary dimensions:** the analyst's toolkit gets bigger, not different

Fill-in / Q&A

The following slides apply when running the demo live or fielding detailed questions about how Genie behaves under the hood.

Genie in Action on the Existing Catalog

Two questions from the base catalog, Genie answering what it's built for:

- **Q1:** "What are the top 10 accounts by total amount spent across merchants?" Clean ranked list; standard aggregation over `transactions`
- **Q2:** "Show accounts with above-average spend and more than 20% of transactions at night." Join and conditional aggregate; correct top-15 with night ratio and balance
- **Genie doing its designed job** on the catalog the customer already runs
- **The SQL shapes are standard:** all dimensions live in the base tables

Deterministic Foundation, Non-Deterministic Translation

- **Genie generates different SQL each run.** Same question, different shape: `RANK()=1` one time, `LIMIT 100` the next. That is how text-to-SQL works
- **GDS columns are fixed.** Same projection, same scores, every time. The signal does not move
- **The combination is reliable:** SQL variance only changes how much signal Genie surfaces, never whether it exists

Defensibility

- **GDS produces features with published mathematical definitions:** PageRank, Louvain, Jaccard
- **Humans and downstream models adjudicate, not the pipeline**
- **The pipeline surfaces candidates; it does not label fraud**

Backup Anchor: Ring Share by Region (Before)

"What share of accounts send more than half their transfer volume to five or fewer repeat counterparties, broken out by region?"

- **Genie's SQL:** per account, rank counterparties by transfer volume; flag accounts sending over half their volume to their top 5
- **Share of accounts flagged:** 95.5% to 96.3% across all six regions
- **Top:** US-East with 96.3% of accounts flagged; **Bottom:** EU-West with 95.5%

A reasonable ranking: every region looks alike, with no minority to triage.

Backup Anchor: Ring Share by Region (After)

"What share of accounts sits in communities flagged as ring candidates, broken out by region?"

- **What changed:** BEFORE flagged counterparty concentration by transfer volume; AFTER filters to accounts whose community is flagged is_ring_candidate
- **Share of accounts in ring communities:** 4.69% to 5.51% across all six regions
- **Top:** US-West with 5.51% of accounts in ring communities; **Bottom:** APAC with 4.69%

Structurally defined minority, roughly a tenth the size of the proxy minority.

Validation: Merchant Ring-Candidate Share (Before)

"Which merchants are most commonly visited by the top 20 accounts by total transaction volume?"

- **Genie's SQL:** rank accounts by total volume, count merchant visits in the top 20
- **243 merchants returned:** no merchant visited by more than 2 of the 20 accounts
- **Top merchant:** Barnes-Harris (retail) with 2 visits; the remaining 242 merchants with 1 visit each
- **Shape:** the cohort is completely dispersed, with no shared merchants and no visible concentration

A dispersed list: no merchant stands out as a triage priority.

Validation: Merchant Ring-Candidate Share (After)

"For James-Conway, Cardenas and Sons, Johnson, Williams and May, and Meyer Ltd, what share of each merchant's customers are members of ring-candidate communities, and how does that compare to the book baseline?"

- **What changed:** BEFORE ranked by volume; AFTER measures ring-candidate share of each merchant's customer base against the ~4% book baseline
- **James-Conway (crypto):** 76.2% ring-candidate share, ~19× above the ~4% baseline
- **Cardenas and Sons (utilities):** ~4%, at baseline
- **Johnson, Williams and May (grocery):** ~4%, at baseline
- **Meyer Ltd (retail):** ~4%, at baseline

Three of four are at baseline. James-Conway is the one outlier worth investigating.

Validation: High-Volume Account Community Membership (Before)

"For the top 20 accounts by total transaction volume, how many unique merchants did each account visit?"

- **Genie's SQL:** rank accounts by volume, count distinct merchants per account
- **Range:** 7 to 21 unique merchants across the top 20
- **Top volume account:** account 13318 ($18,429) visited 9 merchants; account 11764 ($11,267) visited 21
- **No correlation** between transaction volume and number of merchants visited

Looks like legitimate diverse spending. No structural red flag visible.

Validation: High-Volume Account Community Membership (After)

"For accounts in the top 20 by total transaction volume, what is their community membership status and risk tier?"

- **What changed:** BEFORE counted unique merchants; AFTER adds community_id and fraud_risk_tier from Gold
- **19 of 20 accounts:** low risk, concentrated in three communities (16163, 6049, 3040)
- **1 account:** high risk, with account 3404 ($12,996) in community 3040
- **Merchant diversity is confirmed legitimate:** spread across 243 merchants is a community 16163 / 6049 pattern, not a ring pattern

The before interpretation was correct. High volume and diverse merchants is not a layering signal here.